

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 2

BIOLOGY

Paper 1 Multiple Choice

9747/01

19 September 2008

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, class and tutor's name on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

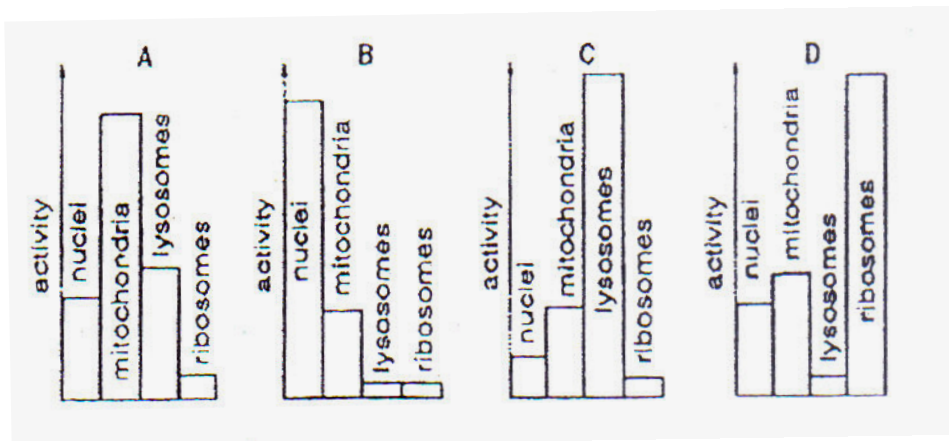
Any rough working should be done in this booklet.

- 1 A microscope is set up to examine the surface of a leaf. The 40x objective lens and a 7x eyepiece is being used. The diameter of the field of view is 0.4mm.

What is the approximate density of the stomata if 15 of them are visible under these conditions?

- A. 33 stomata/mm²
 B. 120 stomata/mm²
 C. 528 stomata/mm²
 D. 1680 stomata/mm²
- 2 A piece of mammalian tissue was homogenized and subjected to differential centrifugation. Four subcellular fractions were investigated to determine their relative biochemical activity. The diagrams indicate the relative activity of certain biochemical processes in these fractions.

Which diagram indicates the fraction with maximum synthesis of messenger RNA?



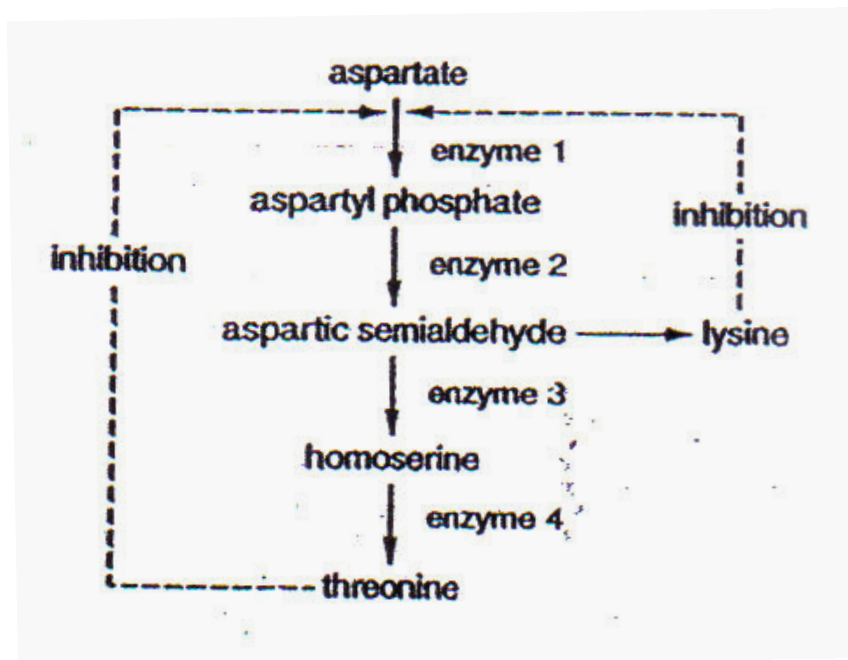
- 3 Which of the following is a characteristic of saturated fats which distinguishes them from unsaturated fats?
- A. They have no double bonds between their carbon atoms.
 B. They are composed only of fatty acids and glycerol.
 C. They contain a high proportion by mass of oxygen.
 D. They contain only unbranched fatty acids.
- 4 The following statements describe three orders of structure of the insulin molecule.
- I. The molecule consists of two polypeptide chains joined and folded around one another.
 II. Each polypeptide chain is of known sequence and number of amino acids.
 III. The amino acids in each chain are coiled into a helix and held in position by hydrogen bonds.

Which order is described by each statement?

	statement I	statement II	statement III
A.	primary	secondary	tertiary
B.	primary	tertiary	secondary
C.	secondary	tertiary	primary
D.	tertiary	primary	secondary

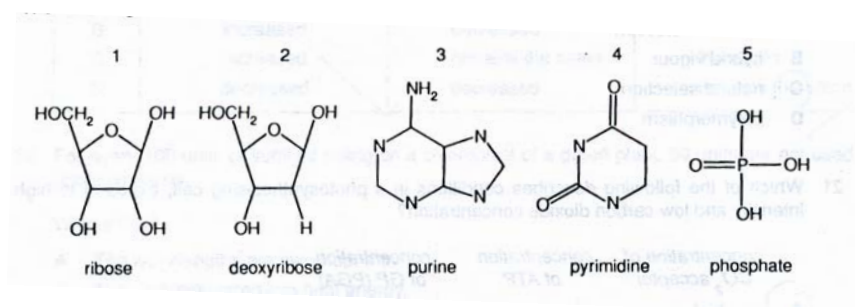
- 5 A culture of bacteria produces the food supplement lysine by the metabolic pathway shown below.

Which change in enzyme activity will result in the greatest increase in lysine yield?



	Enzyme	Change in activity
A	1	Decrease
B	2	Increase
C	3	Increase
D	4	Decrease

6

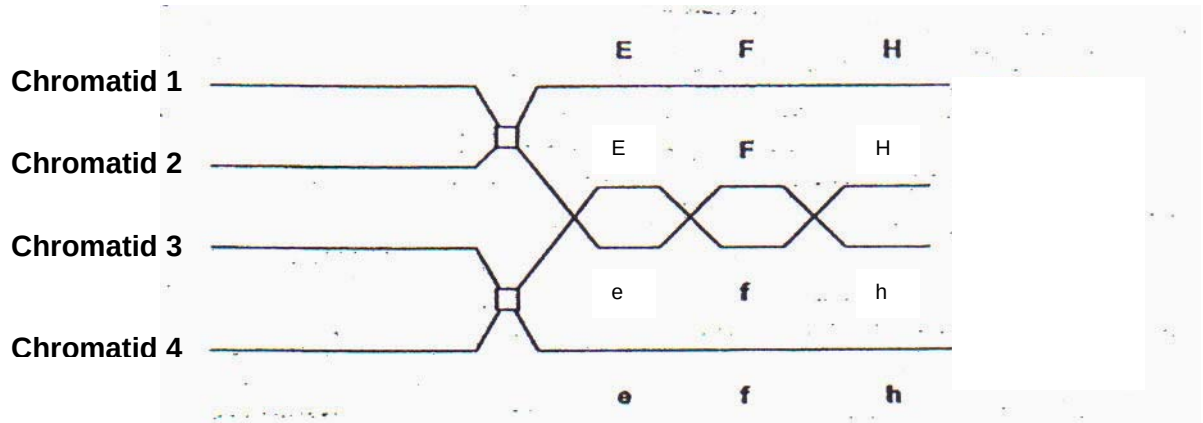


Which of these molecules are linked to form a nucleotide containing uracil?

- A. 2, 3 and 5
- B. 2, 4 and 5
- C. 1, 2 and 5
- D. 1, 4 and 5

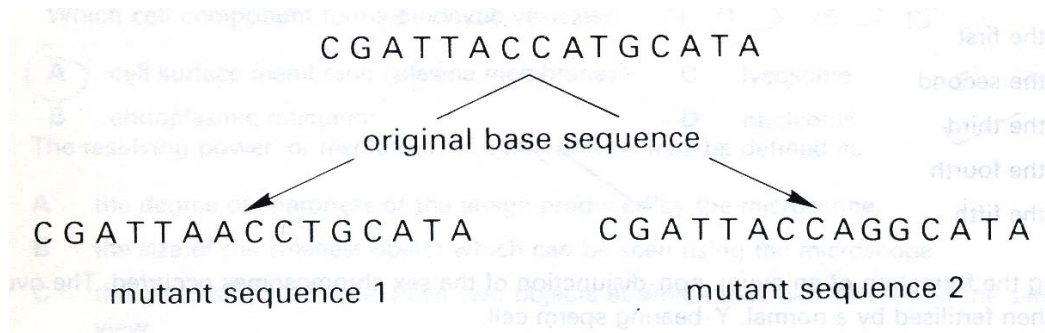
- 7 The diagram below shows a pair of homologous chromosomes during prophase of meiosis. Crossing over took place at all the chiasmata.

Determine the resulting segregation of the named alleles.



- A Chromatid 1: EFH
Chromatid 2: eFh
Chromatid 3: eFH
Chromatid 4: efh
- B Chromatid 1: EFH
Chromatid 2: Efh
Chromatid 3: eFH
Chromatid 4: efh
- C Chromatid 1: EFH
Chromatid 2: eFH
Chromatid 3: Efh
Chromatid 4: efh
- D Chromatid 1: EFH
Chromatid 2: efh
Chromatid 3: EFH
Chromatid 4: efh

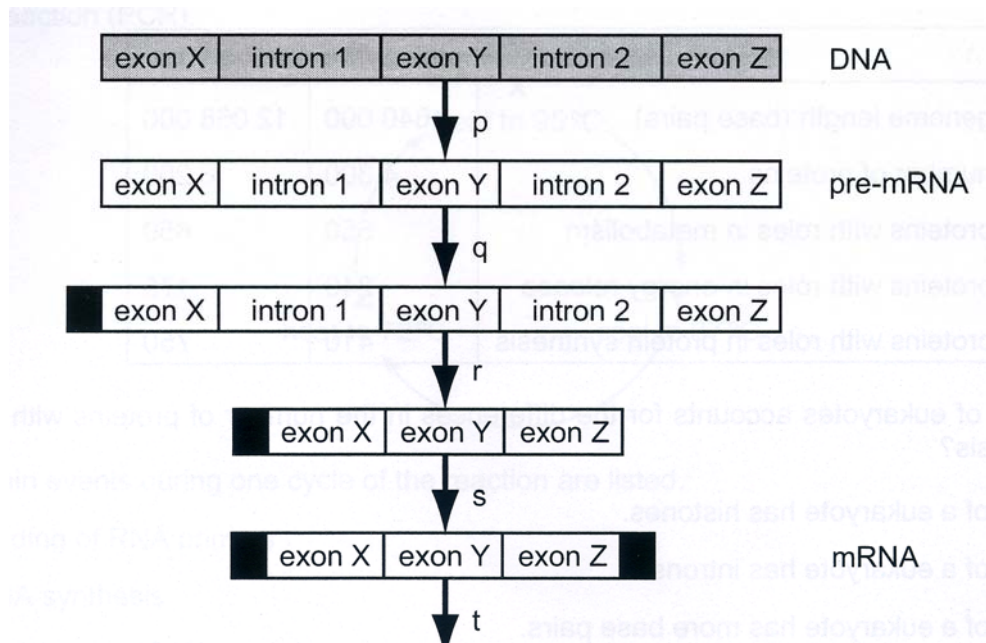
- 8 Cells treated with colchicine often mutate. They fail to produce spindles during cell division. What would be the result of this?
- A chromosome deletion
 - B chromosome translocation
 - C gene deletion
 - D polyploidy
- 9 The diagram shows two types of gene mutation.



What types of gene mutation produce a mutant sequence 1 and 2?

- | | 1 | 2 |
|---|--------------|--------------|
| A | insertion | deletion |
| B | inversion | substitution |
| C | substitution | insertion |
| D | substitution | inversion |
- 10 If the nucleus of a hamster sperm cell contains 3.4 picograms (pg) of DNA, what mass of DNA is the nucleus of an actively dividing hamster skin cell likely to contain at the end of interphase?
- A 3.4 pg
 - B 6.8pg
 - C 13.6 pg
 - D 27.2 pg
- 11 Which mutation could be inherited from only the maternal parent?
- A nuclear DNA
 - B ribosomal DNA
 - C messenger RNA
 - D mitochondrial DNA
- 12 Which of the listed features are characteristic of a prokaryotic genome?
- 1 polycistronic mRNA
 - 2 DNA associated with histones
 - 3 translation by 80S ribosomes
 - 4 group of structural genes clustered together
- A 1 only
 - B 3 only
 - C 1 and 4
 - D 3 and 4

- 13 The diagram represents the processing of pre-messenger RNA.



During the process of pre-messenger RNA each of the listed events occurs.

- 1 capping
- 2 polyadenylation
3. splicing
- 4 transcription
- 5 translation

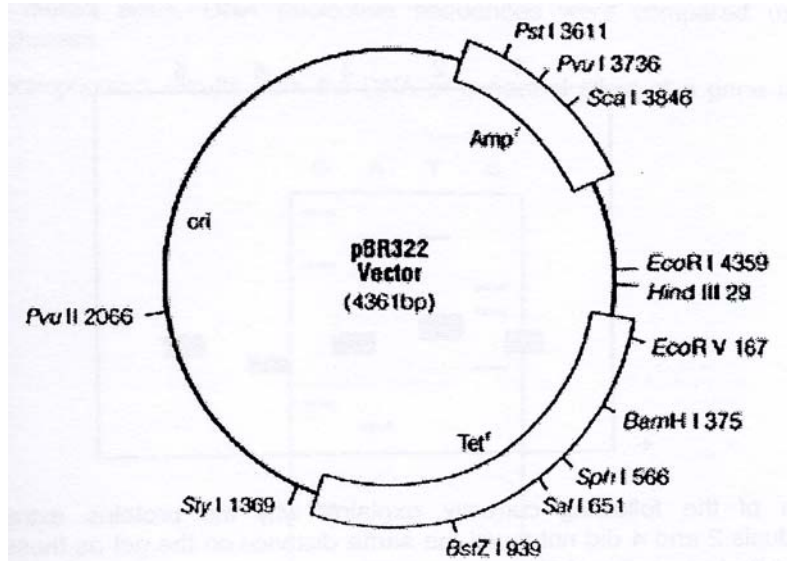
Which correctly identifies the processes p, q, r, s and t?

	p	q	r	s	t
A	5	1	3	2	4
B	4	1	3	2	5
C	5	2	1	3	4
D	4	2	1	3	5

- 14 When glucose is absent and lactose is present, lactose binds to the operon's repressor. Which statement best describes what happens subsequently?

- A Repressor binds to promoter, transcription is initiated.
- B Repressor binds to operator, transcription is not initiated.
- C Repressor fails to bind to structural genes, transcription is not initiated.
- D Repressor fails to bind to the operator, transcription is initiated

- 15 pBR322 vector is used to clone a eukaryotic gene which has been digested by the restriction endonuclease *Bam*HI.



Following transformation, bacterial cells were grown in four different media, as shown below:

- 1 nutrient medium + ampicillin
- 2 nutrient medium + tetracycline
- 3 nutrient medium + without antibiotics
- 4 nutrient medium + ampicillin + tetracycline

Which of the following media would bacterial cells contain the recombinant plasmids grow in?

- A 1 and 2
 - B 1 and 3
 - C 2 and 3
 - D 2 and 4
- 16 Which of the following are the two most common processes that lead to production of multiple functional proteins that are derived from the same DNA sequence?
- A Alternative splicing and protein folding
 - B Alternative splicing and post-translational chemical modifications
 - C Protein folding and post-translational chemical modifications
 - D Transcriptional regulation and post-translational chemical modifications
- 17 Which of the following is a common type of tumor suppressor genes?
- A Genes that increase the rate of the cell cycle or stimulate cell division.
 - B Genes involved in performing DNA synthesis.
 - C Genes involved in maintenance of cell cycle checkpoints.
 - D Genes involved in signal transduction
- 18 During bacterial conjugation mediated by the F-plasmid, the function of the pilus is to
- A synthesize the DNA single stranded prior to transfer.
 - B draw the donor and recipient cells in contact prior to DNA transfer.
 - C facilitate transfer of single-stranded DNA across the cell membrane.
 - D facilitate transfer of double-stranded DNA across the cell membrane.

19 Which of the following statements about eukaryotic control elements are correct?

- i Attachment of RNA polymerase to promoter is achieved by interaction of activator proteins
- ii Attachment of activator proteins to enhancers increases the basal activity of the promoter
- iii Attachment of activator proteins to silencers suppresses the basal activity of promoter
- iv Enhancers and silencers are DNA sequences found upstream of structural genes and can regulate gene expression

- A ii and iv
- B iii and iv
- C i, ii and iv
- D i, iii and iv

20 Red green colour blindness is a sex-linked recessive trait in humans.

P gen $X^R X^r$ x $X^r Y$

F1 gen $X^r Y$ (Dave)

Which of the following represents the genotypes of Dave's grandparents?

	Maternal grandmother	Maternal grandfather	Paternal grandmother	Paternal grandfather
A	$X^R X^R$	$X^R Y$	$X^R X^r$	$X^R Y$
B	$X^R X^r$	$X^r Y$	$X^R X^R$	$X^r Y$
C	$X^R X^r$	$X^R Y$	$X^R X^r$	$X^R Y$
D	$X^R X^R$	$X^r Y$	$X^R X^R$	$X^r Y$

21 In birds, the male is the homogametic sex. A male bird showing the recessive trait was mated with a female showing the dominant trait of a characteristic governed by a pair of alleles which are sex-linked.

What is the probability that the male offspring will show the dominant trait?

- A. 0.25
- B. 0.50
- C. 0.75
- D. 1.00

22 In *Drosophila*, ebony body colour and vestigial wings are recessive to the wild type and are not linked. Crossing two flies heterozygous for both characters produced 256 offspring.

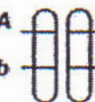
How many offsprings would be expected to have ebony body colour with normal wings?

- A. 16
- B. 48
- C. 96
- D. 144

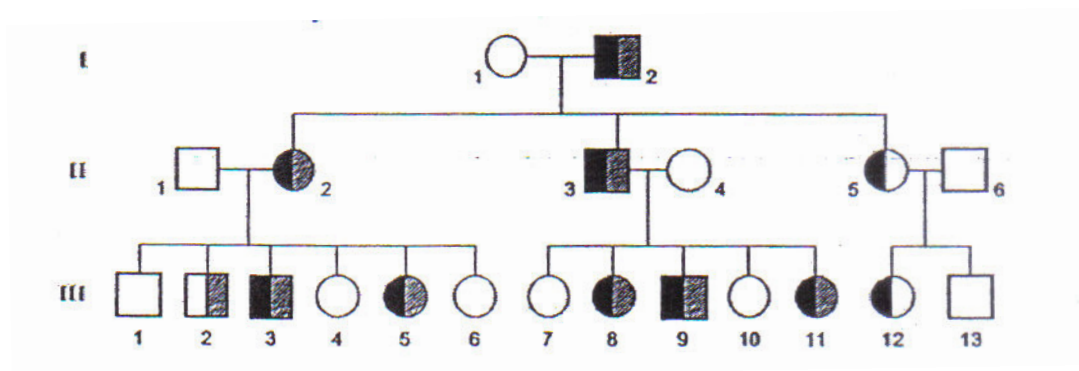
23 Two linked genes in humans have the following alleles.

- Gene 1 A: secretor
 a: non-secretor
- Gene 2 B: production of protein X
 b: no protein X produced

Information regarding these traits in a family was collected and the chromosome composition of each member for these two genes was established. It was found that, in all, there were four different chromosome compositions in the family. The four types are shown in the following table.

	Type one	Type two	Type three	Type four
Genotype				

The pedigree for the family is shown below.



The code for interpreting the symbols is

secretors and produce protein X	non-secretors and produce protein X	secretors and no protein X	non-secretors and no protein X
 	 	 	 

It is reasonable to conclude that the genotype of individual

- A I-1 is type 1.
- B II-2 is type 2.
- C III-2 is type 3.
- D III-12 is type 4.

- 24** A pure breeding plant with red flowers and tall stems was crossed with one with white flowers and short stems. The F₁ plants all had pink flowers and tall stems. These were interbred to produce an F₂ generation. The following ratios of plant phenotypes were obtained in this generation.

3/16	red flowers tall plants
1/16	red flowers short plants
3/8	pink flowers tall plants
1/8	pink flowers short plants
3/16	white flowers tall plants
1/16	white flowers short plants

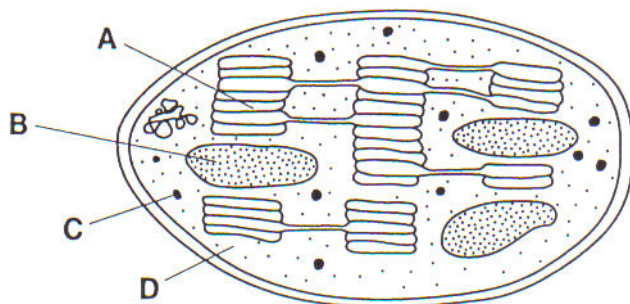
Which of the following statements is true with respect to inheritance of flower colour and plant height?

- A. Red flowers and tall stature are both dominant characteristics.
 - B. White flowers and short stature are both dominant characteristics.
 - C. Red flower colour is incompletely dominant to white flower colour and tall stature is dominant to short stature.
 - D. Red flower colour is incompletely dominant to white colour and tall stature is incompletely dominant to short stature.
- 25** A strain of toad has only one nucleolus in the nucleus of each cell instead of the usual two. When toads with one nucleolus per cell are mated, approximately a quarter of the offspring have two nucleoli per nucleus, half have one nucleolus per nucleus and a quarter have no nucleoli.

What is the most likely explanation of these results?

- A. The possession of one nucleolus is due to autosomal linkage.
- B. The possession of one nucleolus is due to the heterozygous condition.
- C. The allele for the presence of two nucleoli is recessive.
- D. The allele for the presence of two nucleoli is dominant

- 26** The diagram represents a section through a chloroplast.



In which area does chemiosmosis takes place?

- 27** From yeast cells, a mixture of enzymes can be obtained that catalyses the conversion of glucose to carbon dioxide and ethanol. Pyruvate is produced during this process.

Which combination of substances would increase the rate of ethanol production by these enzymes?

- A ATP and lactate
- B ATP and NAD
- C NAD and coenzyme A
- D NAD and lactate

- 28** The concentration of carbon dioxide in a sample of air was found to be 280 ppm (parts per million). A control experiment was designed to measure the concentration of carbon dioxide in the air after it had flowed over the leaves of a green plant. Measurements were taken at a range of light intensities.

The following results were obtained:

Light intensity (% of full sunlight)	Concentration of carbon dioxide in air after flowing over leaves (ppm)
75	253
50	252
25	254
10	280

Which one of the following statements is not consistent with these results?

- A At the lower light intensities tested, the rate of photosynthesis is limited by light intensity.
- B At the higher light intensities tested, the rate of photosynthesis is affected by factors other than light.
- C In the dark, the concentration of carbon dioxide in the air after it had flowed over leaves would be at least 280 ppm.
- D At a light intensity of 10% of full sunlight, photosynthesis does not occur.

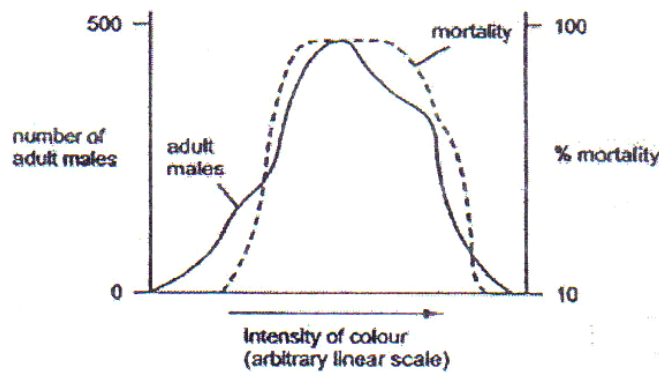
- 29** A person has poorly functioning β cells in the islets of Langerhans. What will be the concentration of glucose, insulin and glucagon in the blood after that person exercised for an hour and having eaten a meal rich in carbohydrates four hours ago?

	glucose	insulin	glucagon
A	high	low	high
B	high	high	low
C	low	low	high
D	low	low	low

- 30 A particular type of autoimmune disease causes antibodies to destroy acetylcholine receptors of neurons.

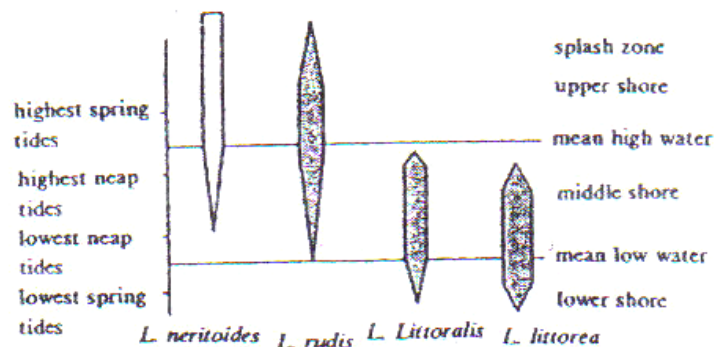
What effect will this have on the functioning of the nervous system?

- A Presynaptic neurons will be unable to release neurotransmitters into the synaptic clefts.
 - B Neurons will be unable to propagate action potentials along their axons.
 - C Depolarized neurons will be unable to reestablish an ionic gradient across their membranes.
 - D Postsynaptic neurons will be unable to detect signals from presynaptic neurons.
- 31 The graph below shows data on a population of a species of moth which shows considerable variation in colour intensity



Which conclusion can be made from this graph?

- A Colour variation is environmentally induced.
 - B Colour variation is genetically determined.
 - C Extreme forms are favoured by natural selection.
 - D The species shows discontinuous variation with respect to colour.
- 32 The diagram below shows the frequency and distribution of four *Littorialis* species (periwinkles) on a rocky shore. All feed in a snail-like manner by grazing on algae.



Which of the following factors could not directly contribute to this distribution pattern?

- A Variation in the tolerance of each species to desiccation.
- B Variation in the tolerance of each species to wave action.
- C Photoperiod and seasonal change in day length.
- D Differential selection of periwinkles by predators

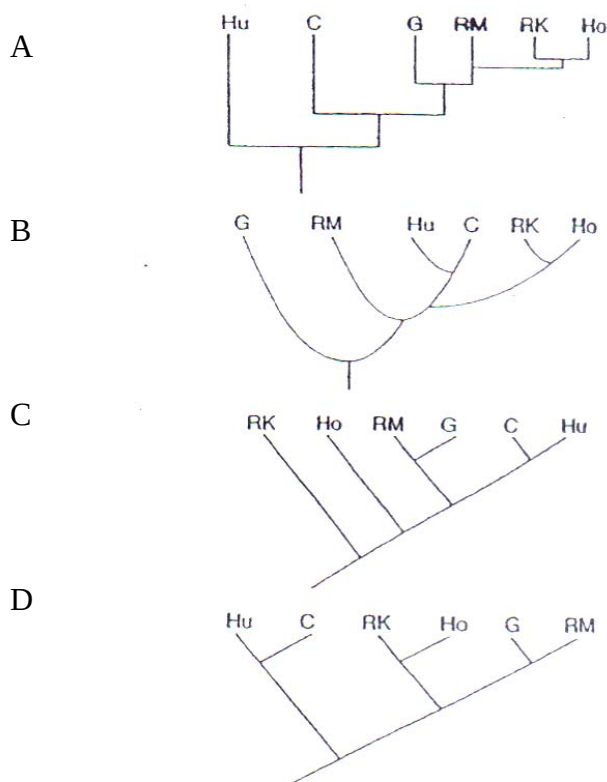
33 What are the major effects of gene flow between subpopulations?

- A Increase genetic variation within and between subpopulations.
- B Decrease genetic variation within and between subpopulations.
- C Decrease genetic variation within but increase genetic variation between subpopulations.
- D Increase genetic variation within but decrease genetic variation between subpopulations.

34 The table below shows the amino acid sequence of part of the beta-haemoglobin molecule found in 6 different vertebrates. The beta-globin chain has a total of 146 amino acids. Only the amino acids found in positions 101 to 116 are represented.

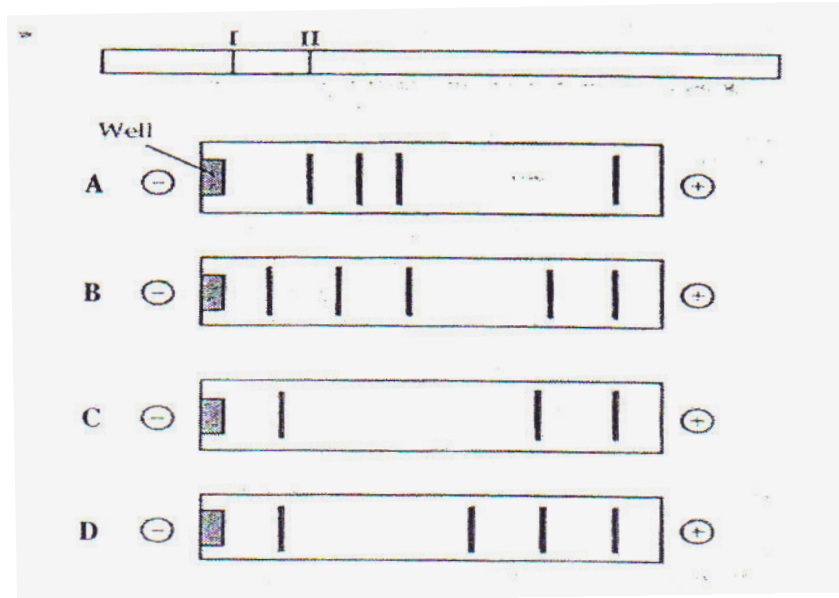
amino acid position	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116
human (Hu)	glu	asn	phe	arg	leu	leu	gly	asn	val	leu	val	cys	val	leu	ala	his
red kangaroo (RK)	glu	asn	phe	lys	leu	leu	gly	asn	ile	ile	val	ile	cys	leu	ala	glu
rhesus monkey (RM)	glu	asn	phe	lys	leu	leu	gly	asn	val	leu	val	cys	val	leu	ala	his
gorilla (G)	glu	asn	phe	lys	leu	leu	gly	asn	val	leu	val	cys	val	leu	ala	his
chimpanzee (C)	glu	asn	phe	arg	leu	leu	gly	asn	val	leu	val	cys	val	leu	ala	his
horse (Ho)	glu	asn	phe	arg	leu	leu	gly	asn	val	leu	ala	leu	val	val	ala	arg

Using the information above, a suitable phylogenetic diagram would be



- 35 The restriction fragment shown below contains a gene whose recessive allele is lethal. The normal allele has restriction sites for restriction enzyme *pstI* at sites I and II. The recessive allele lacks restriction site I. An individual who had a sister with the lethal trait is being tested to determine if he is a carrier of that lethal trait.

Which of the band patterns would be produced on a gel if he is a carrier?



- 36 Nuclei from both the liver and brain cells of a rat are isolated and incubated with highly radioactive RNA precursors. The RNA transcripts synthesised in these nuclei become radioactively labelled.

These radioactive RNA molecules are incubated with a single-stranded DNA segment (cDNA) complementary to an mRNA molecule found in liver-cell cytoplasm but not in brain-cell cytoplasm.

The RNA and DNA are allowed to form RNA/DNA hybrid double helices with the specific cDNA probe. An RNA-digesting enzyme is added to destroy RNA that is not in hybrid form.

Radioactivity in the remaining hybrids is measured as counts per minute. The results are presented in the table below.

Incubation components	Counts per minute
Liver cDNA probe plus primary RNA transcripts from liver-cell nuclei	15,000
Liver cDNA probe plus primary RNA transcripts from brain-cell nuclei	150

Which of the following statements concerning the DNA sequence corresponding to this liver-specific mRNA molecule is best supported by the data?

- A It is transcribed in both brain- and liver-cell nuclei at approximately the same rate.
- B It is not transcribed in either liver- or brain-cell nuclei.
- C It is not transcribed in brain-cell nuclei but is transcribed in liver-cell nuclei.
- D It may be transcribed in both brain- and liver-cell nuclei but cannot be detected by this experimental procedure.

- 37** The hormone insulin binds to the tyrosine kinase receptors and initiates various signal transduction pathway to generate cellular responses. Which of the following shows the correct sequence of events, following the binding of insulin to the receptor?
- i) phosphorylation of tyrosine kinases
 - ii) dimerization of tyrosine kinase receptor
 - iii) activation of enzyme
- A i, ii, iii
 - B iii, ii, i
 - C ii, i, iii
 - D iii, i, ii
- 38** Which of the following statements regarding stem cells is false?
- A One potential side effect of any embryonic stem cell-based therapy is the formation of tumour.
 - B Stem cells are present within various organs of the adult body.
 - C Stem cells can develop into a whole organism when implanted into the womb.
 - D Stem cells can be grown and expanded indefinitely in culture under appropriate culture conditions.
- 39** Expression of a cloned eukaryotic gene in a prokaryotic cell involves many difficulties. The use of mRNA and reverse transcriptase is part of a strategy to solve the problem of _____.
- A. post-transcriptional processing
 - B. post-translational processing
 - C. nucleic acid hybridization
 - D. restriction fragment ligation
- 40** Which statement best describes the characteristic of a pluripotent cell?
- A Cells that can differentiate into embryonic and extra embryonic cell types.
 - B Cells that can differentiate into all the cells found in various tissues of the body.
 - C Cells that can only produce closely related family of cells.
 - D Cells that can produce only one cell type but have the property of self-renewal.

BLANK PAGE